

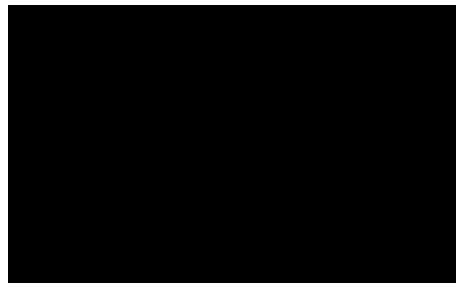
NaviGait: Navigating Dynamically Feasible Gait Libraries Using Deep Reinforcement Learning

Neil Janwani*, Varun Madabushi*, Maegan Tucker

Introduction: Bipedal Locomotion

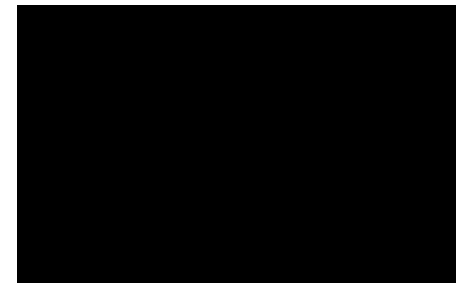
Reinforcement Learning

Adaptable, robust, and highly capable robot control, but depends on *well-defined rewards* or *high-quality demonstrations*



Trajectory Optimization

Constructs appealing motion plans with physically meaningful constraints, yet is brittle and too slow to compute online.



Can we combine the specifiability and natural motion from trajectory optimization with the robust performance of RL?

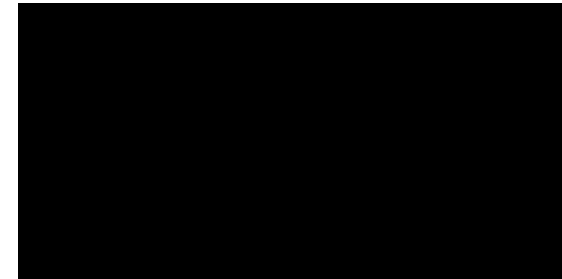
Gait Libraries: Interpolating Between Motion Plans

Possible Approach: transition between different motion plans $\{\mathcal{X}_i\}$ in a gait library $\mathcal{G}$ to stabilize the robot.

1. Physically meaningful motion specification for each $\mathcal{X}_i$.
2. Tune control laws to transition to from current $\mathcal{X}_j$ to new gait $\mathcal{X}_k$.



While this approach can work well, it requires significant heuristics and tuning.

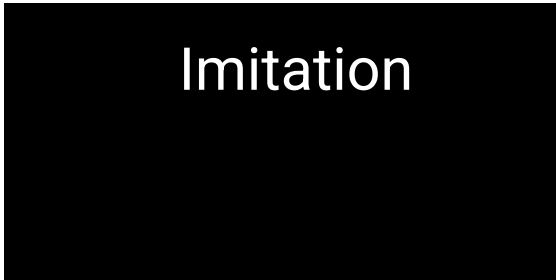


Imitation Learning: Direct Emulation of a Motion Plan

Another Approach: train policy $\pi_{\theta}(a \mid s, \mathcal{X})$ imitate motion plan $\mathcal{X}$ from trajectory optimization.

1. Easy to define reward R for $\mathcal{X}$.
2. (Locally) natural + robust to $\mathcal{X}$

However, when should π_{θ} deviate from the desired motion plan $\mathcal{X}$?



Imitation

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Imitation

Our approach

Approach: NAVIGAiT

NAVIGAiT uses RL to 'fill the gaps' of gait libraries.

1. Synthesize underlying gait library $\mathcal{G}$ via trajectory optimization.
2. Use RL to transition between and correct motion plans in $\mathcal{G}$.

Replaces *transition control* by
retrieving and interpolating $\mathcal{X}_i \in \mathcal{G}$.

Replaces *regulators* by providing
joint-level corrections to gait $\mathcal{X}_i$.

Gait Libraries: Hybrid-Zero Dynamics

Utilize FROST to generate gait $X(\alpha)$ composed via Bézier polynomials.

Bézier polynomials are parametrized by *control points* α .

Gait characteristics like *step width* are included in additional constraints.

$$\begin{aligned} \text{Joint Trajectories} \\ \overbrace{X(\alpha)^*} &= \underset{X(\alpha)}{\operatorname{argmin}} \overbrace{\mathcal{J}(X(\alpha))}^{\text{Cost}} \\ \text{s.t.} \quad &\text{Closed Loop Dynamics} \\ &\text{Impact Invariance} \\ &\text{Physical Feasibility} \\ &\text{Additional Constraints} \end{aligned}$$

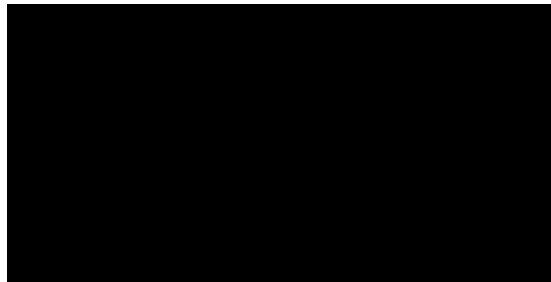
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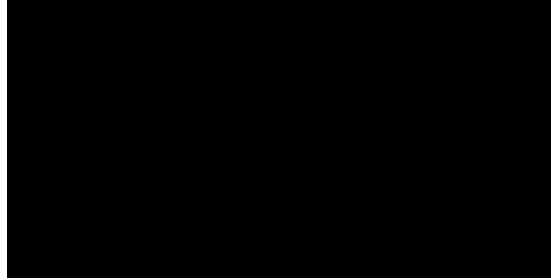
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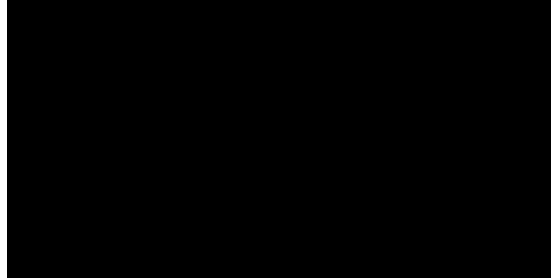
Gait Libraries: Blending and Interpolating



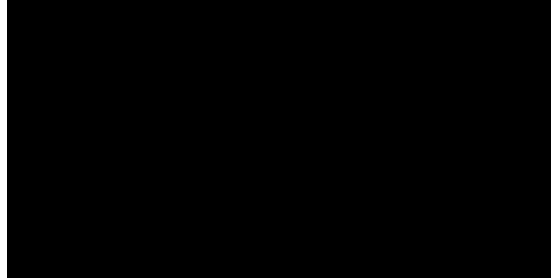
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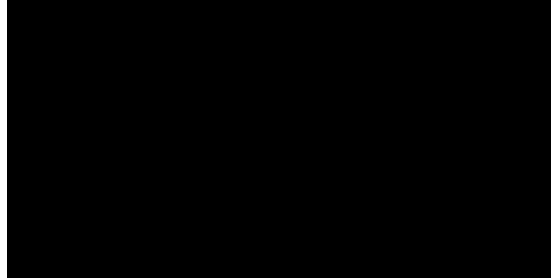
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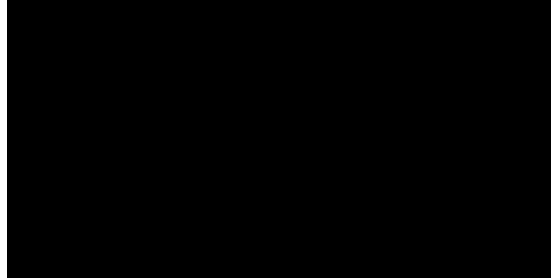
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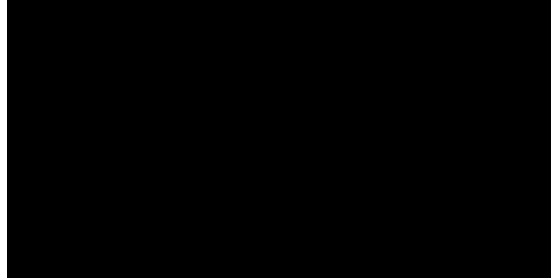
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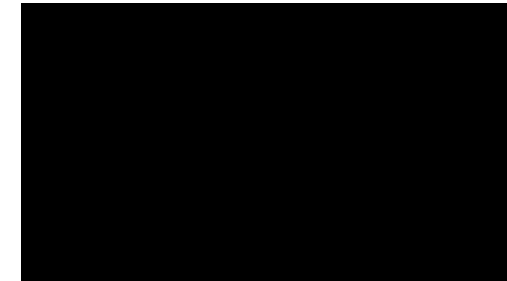
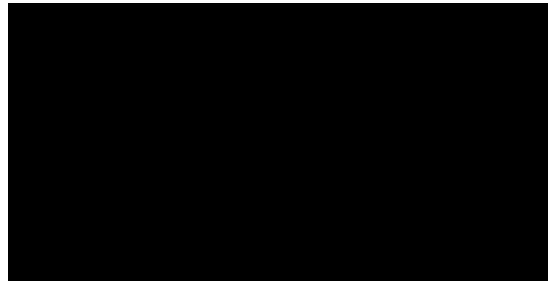




Residual RL: Adapting to Disturbance

We utilize RL to concurrently achieve two objectives:

1. Transition between $\mathcal{X}_i \in \mathcal{G}$ via Δv .
2. Correct the current $\mathcal{X}_i$ via Δq .



Reward Function

$$R_{\text{NaviGait}}^t(s, a) = R_{\text{Gait}}^t + R_{\text{Base Cartesian}}^t + R_{\text{Base Orientation}}^t + R_{\text{torque}} + R_{\Delta v \text{ size}}^t + R_{\Delta v \text{ rate}} + R_{\Delta q \text{ rate}}.$$

Observation (histories)

1. Sensors: IMU + encoders.
2. Reference gait selection.
3. Previous π_θ outputs.

BRUCE Humanoid Robot



We utilize the Westwood Robotics BRUCE Humanoid Robot



Desired joint $\hat{q}_d$ are tracked via low-level PD-control.

We improve simulation fidelity by including

1. Differential drive mechanism (hip)
2. 4-bar linkage mechanism (knee-ankle)

Results: Training and Basic Performance

NAViGAiT learns core walking skills, and emergent turning behavior + works on hardware!



We also compare NaviGAiT against two baselines:
Canonical RL and **Imitation RL**.

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NAViGAiT learns locomotion skills faster than both baseline algorithms.



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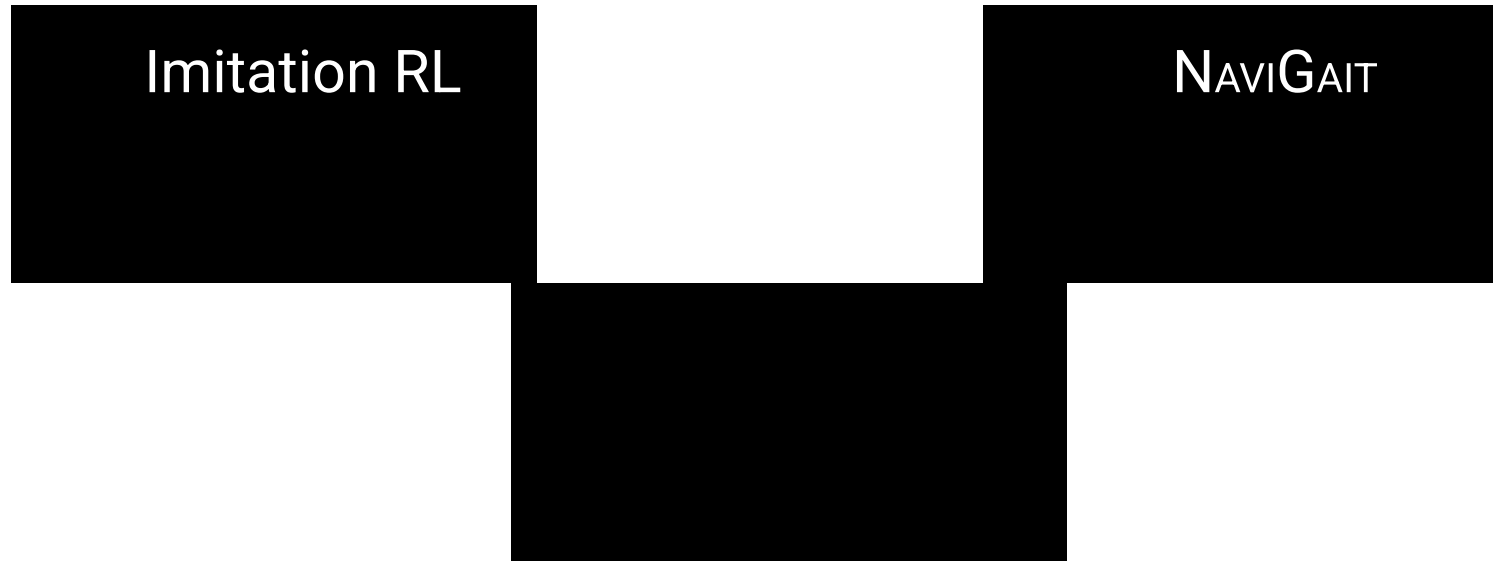
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Canonical RL and **Imitation RL**.

NAViGAiT learns similar robustness to **Imitation RL**.



Results: Imitation Accuracy

We demonstrate that NAVIGAiT has improved imitation accuracy.



Conclusion and Future Work

We present NAVIGAiT, a framework for applying RL to strategically transition between motion plans within a gait library.

1. Learns core locomotion skills faster than baseline approaches
2. Achieves similar robustness as Imitation RL.
3. Demonstrates superior imitation accuracy over Imitation RL.

However, NAVIGAiT has its limitations, paving way for future work

1. Foot cross-over skills
2. Discouraged from extreme maneuvers

Thank You!

Questions?